frontiers

Supplementary Material

Table 1. Studies evaluating sargramostim for hematopoietic recovery after myelosuppressive therapy

Citation	Design/Patient Population	Treatment	Efficacy	Treatment Adverse Events Specific to Sargramostim or Comparator	Comment
Fields 1994 (1)	Phase 2 Hematologic (n=22; lymphoma: n=19) and solid malignancy (n=93; breast cancer n=80)	ICE then ABMT; Sargramostim 250 µg/m^2/d IV over 2h (max 21d) (n=47) vs G-CSF 10 µg/kg/d IV over 30min (n=24) vs no CSF (n=36)	*Sargramostim vs G-CSF vs no CSF:* • Median time to ANC >500/µL: 18d vs 16d vs 21d (p = .001[b]) • Median time to ANC >1000/µL: 21d vs 19d vs 24d (p = NR)	None reported	*With sargramostim:* • Decreased time to ANC >500/µL compared to no CSF
O'Day 1994 (2)	Phase 2; Open label Hematologic malignancy (N=72 lymphoma)	ABMT preparative regimen[c] then Sargramostim 250 µg/m^2/d continuous IV x 21d, started within 2h of marrow reinfusion (n=26) vs historic controls (n=46)	*Sargramostim vs control:* • Median time to ANC ≥500/µL: 14d vs 22d (p = .0001) • Median time to ANC ≥1,000/µL: 16d vs 27d (p < .0001) • Median time to platelets ≥20,000/µL: 22d vs 19d (p = NS) • Median time on IV antibiotics: 7d vs 13d (p = .0004) • Median duration of hospitalizations: 24d vs 28d (p = .0003)	*TRAE sargramostim vs control:* • Fever: ○ >101°F during transplant: 96% vs 83% ○ During agranulocytosis: 48% vs 95% ○ Persistent fever (ANC >500/µL), unrelated to infection: 72% vs 11% • Rash: 42% vs 41% • Serositis: 12% vs 0% • Peripheral edema: 23% vs 4% • Bone pain: 15% vs 0%	*With sargramostim:* • Decreased time to ANC ≥500 and 1,000/µL • Fewer antibiotic days • Reduced length of hospital stay

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Bunn 1995 (3)	Phase 3; Randomized; Open label Multicenter Solid malignancy (N=230 small cell lung cancer)	Chemo-radiotherapy[d] then: No sargramostim vs Sargramostim 250 μg/m^2 SC BID d4-18 in 21d cycles x 6	*Sargramostim vs no sargramostim:* • ANC <500/μL: 18% vs 24% (p = .01) • Platelets <25,000/μL: 35% vs 6% (p < .001)	*Sargramostim vs no sargramostim:* • Toxic deaths: n = 9 (n = 5 pulmonary) vs n= 1 (pulmonary) (p < .01) • Grade ≥3 non-hematologic TEAE higher with sargramostim: o Infection (p = .04) o Respiratory infection (p < .001) o Dyspnea (p < .01)	*Sargramostim vs no sargramostim:* • Increased thrombocytopenia • Increased toxic deaths and non-hematologic AE • Avoid sargramostim prophylaxis in small cell lung cancer therapy with cisplatin, etoposide and thoracic radiation
Jones 1996 (4)	Phase 3; Randomized; Double-blind; Placebo-controlled Solid malignancy (N=142 breast cancer)	Chemotherapy[e] then: Placebo vs Sargramostim 250 μg/m^2 SC d3-15 in 21d cycles x 4	*Sargramostim vs placebo:* • Median days ANC <500/μL: 2.8d vs 6.8d (p < .001) • Median days ANC <1,000/μL: 6.0d vs 9.1d (p < .001)	*TEAE sargramostim vs placebo:* • Late-cycle grade 4 thrombocytopenia (cycles 3 and 4) in GM-CSF recipients • Grade 1-2 injection site reactions 53% vs 14% • Low-grade fever: 51% vs 26% • No difference in any Grade 3-4 nonhematologic TEAE	• GM-CSF use associated with significant increase in FAC dose-intensity as chemotherapy delivered on time more often in GM-CSF group (p < .001 for each drug) • Decreased days of ANC <500 and 1,000/μL with sargramostim
O'Shaughnessy 1996 (5)	Phase 3; Open label; Randomized Solid malignancy (N=53 breast cancer)	FLAC[f]; starting d4 until ANC >5,000/μL and platelets >100,000/μL: Sargramostim 250 μg/m^2/day SC vs PIXY321[g] 375 μg/m^2 BID SC	*Sargramostim vs PIXY321:* • Mean days ANC <500/μL: 4.5d vs 5.6d (p = NS) • Mean days ANC <1000/μL: 7d vs 8.3d (p = .015)	*TRAE with sargramostim:* • Grade 3 fatigue (n = 1) *Sargramostim vs PIXY321:* Grade 2-3 local skin reactions: 6 vs 21 (p < .001)	Fewer days ANC <1,000/μL and fewer skin reactions with sargramostim vs PIXY321

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Yau 1996 (6)	Phase 3 Randomized; Double-blind; Placebo-controlled; Multicenter Hematologic (n=10 lymphoma) and solid malignancy (n=46 breast cancer)	Chemotherapy[h] then: Placebo vs Sargramostim 250 µg/m^2/d SC BID d4 until ANC >1,500/µL x 2d (or maximum 49d)	*Sargramostim vs Placebo (cycle 1):* • Median days ANC<500/µL 10d vs 12d (p = .01) • Median days ANC <1,000/µL 10d vs 13d (p = .004)	No differences in grade 3-4 nonhematologic TEAEs	Sargramostim use decreased days ANC <500/µL and 1,000/µL
Beveridge 1997 (7)	Phase 3 Randomized; Double-blind; Multicenter Hematologic (n=27; lymphoma n=23); Solid malignancy (n=84; breast cancer n=58)	Sargramostim 300 µg (193 µg/m^2) (n=75) vs G-CSF 481 µg (7 µg/kg) (n=62) SC QD x 5-7d	Not reported	*Sargramostim vs G-CSF:* • Grade 1 fever: 48% vs 26% (p = .01) • No difference in grade 3 systemic or local adverse events	No statistically significant differences in incidence or severity of local or systemic AEs possibly related to sargramostim vs G-CSF
Beveridge 1998 (8)	Phase 3: Randomized; Double-blind; Multicenter Hematologic (n=50; lymphoma); Solid malignancy (n=90; 50% breast, 32% lung), and other malignancy (n=41)	Sargramostim 250 µg/m^2 SC daily (n=79) vs G-CSF 5 µg/kg SC daily (n=102)	*Sargramostim vs G-CSF:* • Mean time to ANC >1500/µL: 5.7d vs 4.6d (p = .0001) • Mean time to ANC >1000/µL: 5.1d vs 4.5d (p = .009) • Mean time to ANC >500/µL: 3.3d vs 3.6d (p = NS)	*TRAEs of sargramostim vs G-CSF:* • Chills: 1.2% vs 1.9% (p = NS) • Grade 2 Fever, 4-hr post injection: 3.8% vs 10.7% (p = NS)	No difference in time to ANC >500/µL but increased time to ANC >1,000 and 1,500/µL with sargramostim
Schiller 1998 (9)	Phase 2; Multicenter; Single arm Hematologic malignancy (N=55 multiple myeloma)	Autologous CD34$^+$-selected PBSCT plus Sargramostim 500 µg QD IV/SC until ANC >1,000/µL x 3d	*Sargramostim:* • Median time to ANC >500/mL: 12d • Median time to platelets>20,000/mL: 12d	None reported	Effective use of sargramostim with autologous CD34$^+$-selected PBSCT

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Steward 1998 (10)	Phase 3 Randomized; Double-blind; Placebo-controlled; Multicenter Solid malignancy (N=300 small cell lung cancer)	Intensified chemotherapy[i] every 3 wk x 6 cycles vs Standard chemotherapy[i] every 4 wk x 6 cycles Second randomization: Placebo vs Sargramostim 250 µg/m^2 SC daily x 14d between cycles	*Sargramostim vs placebo:* • Incidence febrile neutropenia: ○ Intensified therapy: 56% vs 55% ○ Standard therapy: 53% vs 51%	*TRAE sargramostim vs placebo:* • Injection-site reactions: 5% vs 0% • Arthralgia/myalgias: 1.3% vs 1.3%	Sargramostim use did not significantly decrease time to hematopoietic recovery or incidence of chemotherapy-related adverse events
Le Cesne 2000 (11)	Phase 3 Randomized; Multicenter Solid malignancy (N=314, soft tissue sarcoma)	Standard dose chemotherapy[j] every 3 wk vs High-dose chemotherapy[j] + sargramostim 250 µg/m^2 SC d3-16 or until ANC >1,000/µL every 3 wk	*Standard dose chemotherapy vs high-dose chemotherapy + sargramostim:* • Median progression-free survival 19 wk vs 29 wk (p = .03)	*TEAE Standard dose chemotherapy vs high-dose chemotherapy + sargramostim:* • Febrile neutropenia: 4.6% vs 16.6% (p = .0004) • Grade 3-4 thrombocytopenia 8% vs 50% • Grade 3-4 neutropenia 92% vs 90% • Grade 3-4 asthenia 4.5% vs 16% (p = .0005) • Grade 3-4 stomatitis 3.9% vs 13% (p = .008)	Sargramostim facilitated dose-intensity and improved progression-free survival but no difference in overall survival

[b]p value for sargramostim and G-CSF group vs no CSF group; [c]cyclophosphamide x 2 d + total body irradiation 1,200 cGy (200 cGy BID over 3 d); [d]cisplatin and etoposide d1-3 and thoracic radiation therapy 4,500 cGy (25 fractions); [e]fluorouracil, doxorubicin, and cyclophosphamide on d1; [f]FLAC on d1-3 in 21-day cycles x 5; [g]PIXY321 is an IL-3/GM-CSF fusion protein; [h]cyclophosphamide d1-2, etoposide d1-3, cisplatin d1-3 (maximum 3 cycles); [i] V-ICE (ifosfamide, carboplatin, and etoposide d1-2, then etoposide d3, vincristine d15); [j]doxorubicin and ifosfamide d1

Abbreviations: ABMT, autologous bone marrow transplant; AE, adverse event; ANC, absolute neutrophil count; BID, twice daily; CSF, colony-stimulating factor; FLAC, 5-fluorouracil, calcium leucovorin, doxorubicin and cyclophosphamide; G-CSF, granulocyte colony-stimulating factor; GM-CSF, granulocyte-macrophage colony-stimulating factor; ICE, ifosfamide, carboplatin, etoposide; NS, not significant; PBSCT, peripheral blood stem cell transplant; SC, subcutaneous; TEAE, treatment-emergent adverse event; TRAE, treatment-related adverse event; V-ICE, vincristine, ifosfamide, carboplatin, etoposide.

Supplementary References

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